let us all hope that it is right. What a shame it would be if we had to go and dismiss all the relativists again."

Lectures went on almost continuously from 8:30 in the morning until 6 in the evening with an hour out for lunch, plus 6 P.M. until typically 2 A.M. for informal discussions and arguments. Slipped in among the lectures was a short, ten-minute presentation by a young New Zealander mathematician, Roy Kerr, who was unknown to the other participants. Kerr had just discovered his solution of the Einstein field equation—the solution which, one decade later, would turn out to describe all properties of spinning black holes, including their storage and release of rotational energy (Chapters 7 and 11); the solution which, as we shall see below, would ultimately become a foundation for explaining the quasars' energy. However, in 1963 Kerr's solution seemed to most scientists only a mathematical curiosity; nobody even knew it described a black hole—though Kerr speculated it might somehow give insight into the implosion of rotating stars.

The astronomers and astrophysicists had come to Dallas to discuss quasars; they were not at all interested in Kerr's esoteric mathematical topic. So, as Kerr got up to speak, many slipped out of the lecture hall and into the foyer to argue with each other about their favorite theories of quasars. Others, less polite, remained seated in the hall and argued in whispers. Many of the rest catnapped in a fruitless effort to remedy their sleep deficits from late-night science. Only a handful of relativists listened, with rapt attention.

This was more than Achilles Papapetrou, one of the world's leading relativists, could stand. As Kerr finished, Papapetrou demanded the floor, stood up, and with deep feeling explained the importance of Kerr's feat. He, Papapetrou, had been trying for thirty years to find such a solution of Einstein's equation, and had failed, as had many other relativists. The astronomers and astrophysicists nodded politely, and then, as the next speaker began to hold forth on a theory of quasars, they refocused their attention, and the meeting picked up pace.

The 1960s marked a turning point in the study of radio sources. Previously the study was totally dominated by observational astron?mers—that is, optical astronomers and the radio-observing experimental physicists, who were now being integrated into the astronomical community and called *radio astronomers*. Theoretical astrophysicists, by contrast, had contributed